Probability of Informed No-Tradings: A Copula-Based PIN Model with Zero-Inflated Poisson Distributions

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Outline

Introduction

- The probability of informed trading (PIN) models is widely used to determine the information asymmetry in the security market.

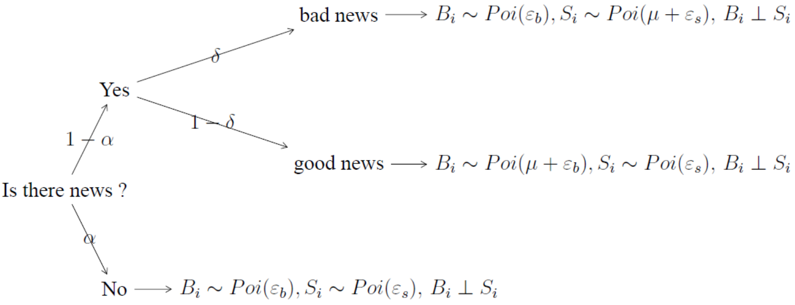


Figure: The structure of the PIN model

Introduction

Regard the classical PIN model as a mixture model:

$$\begin{aligned} f(B,S|\theta) &= (1-\alpha)e^{-\varepsilon_b}\frac{\varepsilon_b^B}{B!}e^{-\varepsilon_s}\frac{\varepsilon_s^S}{S!} \\ &\quad + \alpha\delta e^{-(\mu+\varepsilon_b)}\frac{(\mu+\varepsilon_b)^B}{B!}e^{-\varepsilon_s}\frac{\varepsilon_s^S}{S!} \\ &\quad + \alpha(1-\delta)e^{-\varepsilon_b}\frac{\varepsilon_b^B}{B!}e^{-(\mu+\varepsilon_s)}\frac{(\mu+\varepsilon_s)^S}{S!}, \end{aligned}$$

where $\theta = (\alpha, \delta, \mu, \varepsilon_b, \varepsilon_s)$.

For parameter estimation, using the EM algorithm.

$$\hat{\theta}_{\mathsf{MLE}} = \underset{\Theta}{\operatorname{argmax}} \sum_{i=1}^{n} L(\theta|B_i, S_i)$$

Introduction

- The model assumption is that the correlation between buys and sells is independent $\longrightarrow$ not reasonable in empirical.

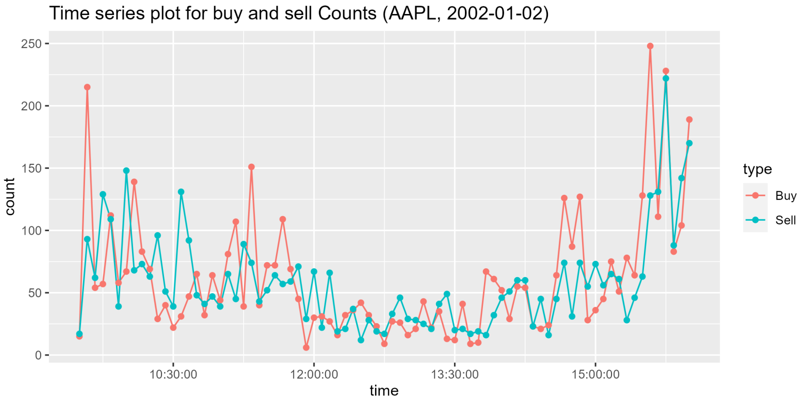


Figure: Time series plot for buy and sell counts (AAPL, 2002-01-02)

Introduction

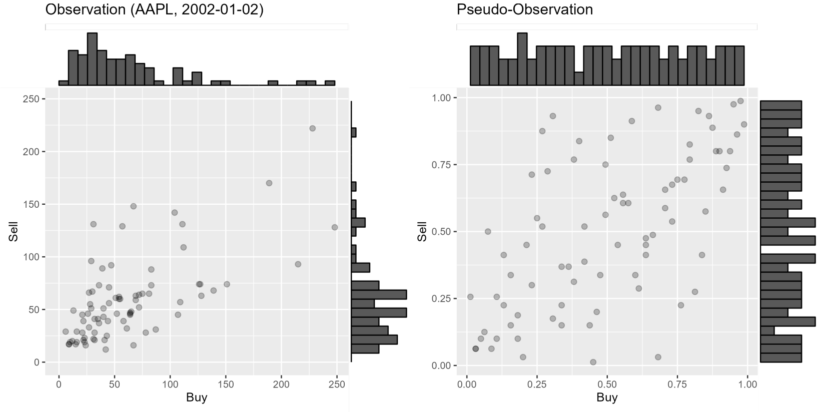


Figure: The scatter plot for buy and sell counts (AAPL, 2002-01-02)

Q: How to evaluate the PIN model with the correlation structure?

Copula : Sklar's theorem

- A copula is a multivariate cumulative distribution function with uniform marginal p.d.f of each variable is on $[0, 1]$.
- Copulas allow us to model marginal distribution and joint relationships separately.

Sklar's theorem

Every multivariate c.d.f. H can be expressed in terms of its marginals F_i's and a copula function C.

If the copula function has parameters $\theta \in \Theta$, and the marginal c.d.f. F_i have parameters $\gamma_i \in \Gamma_i$,then

$$H(\mathbf{x}; \theta, \gamma) = C_\theta(F_1(x_1; \gamma_1), \cdots, F_p(x_p; \gamma_p)),$$

where $\mathbf{x} = (x_1, \cdots, x_p)$

Example of copula

Some examples of copula :

1. Normal copula

$$C_R^{(N)}(u_1, \cdots, u_p) = \Phi_R\left(\Phi^{-1}(u_1), \cdots, \Phi^{-1}(u_p)\right),$$

   where R is a correlation matrix

2. Gumbel copula

$$C_\theta^{(G)}(u_1, \cdots, u_p) = \exp\left[-\left(\sum_i (-\log u_i)^\theta\right)^{1/\theta}\right], \theta \in [1, \infty)$$

3. Clayton copula

$$C_\theta^{(C)}(u_1, \cdots, u_p) = \left[\max\left\{\sum_i (u_i^{-\theta} - 1) + 1, 0\right\}\right]^{-1/\theta},$$

   where $\theta \in [-1, \infty) \backslash \{0\}$

4. Independent copula

$$C^{(I)}(u_1, \cdots, u_p) = u_1 \cdots u_p$$

co-PIN model

The following is the structure of the co-PIN model :

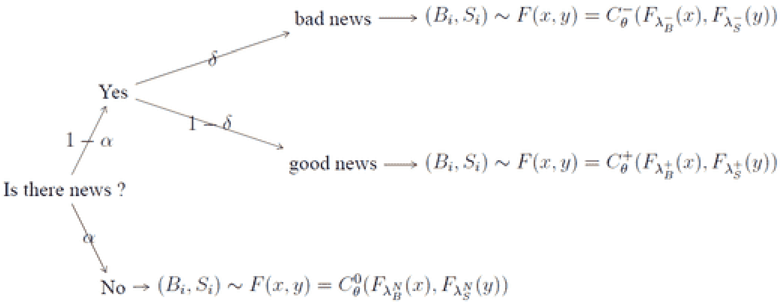


Figure: The structure of the co-PIN model

Density express by copula

- If marginals F_j are continuous, and the density of copula exists, then the multivariate p.d.f. h is given by

$$h(\mathbf{x};\theta,\gamma)=c_\theta(F_1(x_1;\gamma_1),\cdots,F_p(x_p;\gamma_p))\prod_{t=1}^{p}f_t(x_t;\gamma_t),$$

where $c_\theta(u_1,\cdots,u_p)=\frac{\partial^p C_\theta(u_1,\cdots,u_p)}{\partial u_1\cdots\partial u_p}$ is the density of the copula.

- For the discrete marginal, we can express h as the following

$$h(\mathbf{x};\theta,\gamma)=\sum_{i_1=0}^{1}\cdots\sum_{i_p=0}^{1}(-1)^{i_1+\cdots+i_p}$$
$$\times C_\theta(F_1(x_1-i_1;\gamma_1),\cdots,F_p(x_p-i_p;\gamma_p))$$

ECM algorithm

The expectation conditional maximization (ECM) algorithm is the variants of the EM algorithm, which separate the M-step into a sequence of conditional maximization (CM) steps.

Let $\psi = (\theta, \gamma) \in \Theta \times \Gamma$. Divide the M-step into 2 CM-steps :

1. CM-step1 (maximize w.r.t. γ) :

$$\gamma^{(l+1)} = \underset{\Gamma}{\operatorname{argmax}} \sum_{j=1}^{m} \sum_{i=1}^{n} w_{ij}^{(l+1)} \log h_j(\mathbf{x}_i; \theta_j^{(l)}, \gamma_j).$$

2. CM-step2 (maximize w.r.t. θ) :

$$\theta^{(l+1)} = \underset{\Theta}{\operatorname{argmax}} \sum_{j=1}^{m} \sum_{i=1}^{n} w_{ij}^{(l+1)} \log h_j(\mathbf{x}_i; \theta_j, \gamma_j^{(l+1)}).$$

ECM algorithm

1. E-step: Calculate
$$w_{ij}^{(l+1)} = \frac{\pi_j^{(l)} h_j(\mathbf{x}_i; \psi_j^{(l)})}{\sum_{j=1}^m \pi_j^{(l)} h_j(\mathbf{x}_i; \psi_j^{(l)})} \, (i = 1, \cdots, n, \; j = 1, \cdots, m)$$
2. M-step (for $j = 1, \cdots, m$)
   1. Set $\pi_j^{(l+1)} = \frac{1}{n} \sum_{i=1}^n w_{ij}^{(l+1)}$
   2. CM-step1 : $\gamma_j^{(l+1)} = \operatorname*{argmax}_{\Gamma} \sum_{i=1}^n w_{ij}^{(l+1)} \log h_j(\mathbf{x}_i; \theta_j^{(l)}, \gamma_j)$.
   3. CM-step2 : $\theta_j^{(l+1)} = \operatorname*{argmax}_{\Theta} \sum_{i=1}^n w_{ij}^{(l+1)} \log h_j(\mathbf{x}_i; \theta_j, \gamma_j^{(l+1)})$
3. Stop if $\left\| \ell\left(\mathbf{x}; \psi^{(l+1)}, \pi^{(l+1)}\right) - \ell\left(\mathbf{x}; \psi^{(l)}, \pi^{(l)}\right) \right\| < \varepsilon$

Simulation

1. Generate datasets with 500 two-dimensional data points with 2 clusters and then fit the models.
2. Construct the data from specified marginals (Poisson) and copulas (normal, Gumbel, Clayton), then fit the simulation data with the same model setting.
3. Repeat 1000 times, Check the difference with a histogram, and scatter plot of error rates vs. distances between two clusters.
4. To compare different copulas parameters, we use the association measure Kendall's tau,

$$\tau(X_1, X_2) = 4 \iint_{[0,1]^2} C(u,v)\, dC(u,v) - 1$$

Simulation

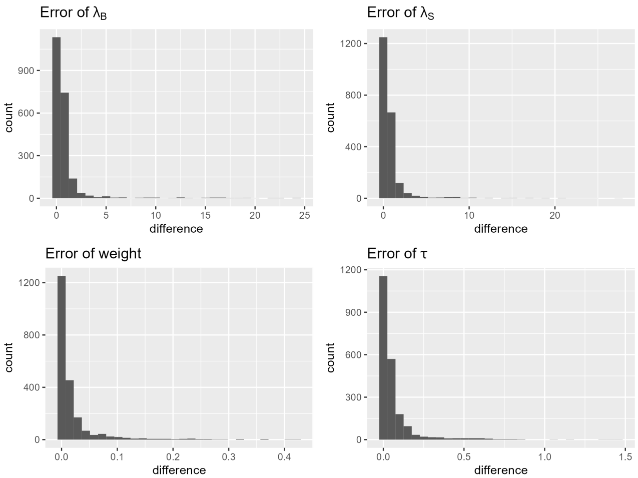


Figure: Histogram of absolute error of parameters

Simulation

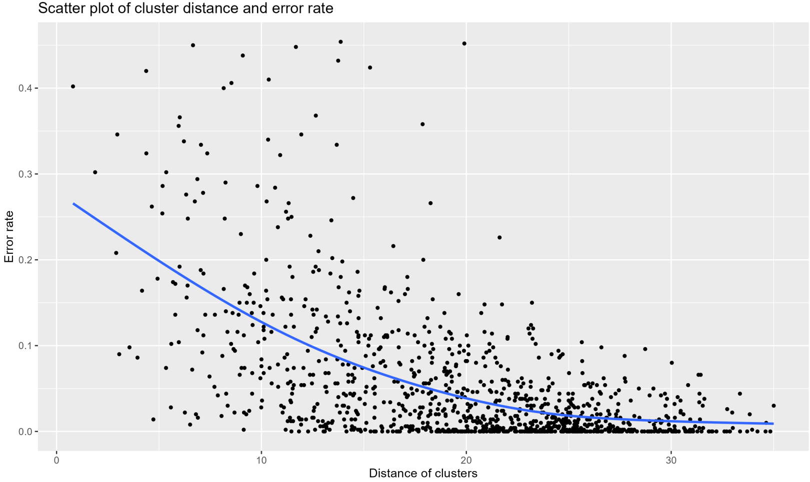


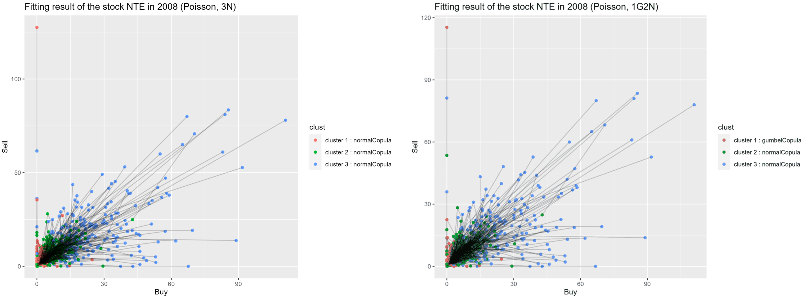
Figure: Scatter plot of classification error rate vs. cluster distance

Real data analysis

- Use the daily trading records of AAPL (large) and NTE (small) stocks in 2008, which were collected every 5 minutes. We only use data within U.S. stock market trading hours, resulting in about 249 days, and 78 data points per day.
- Using zero-inflated Poisson (ZIP) for marginal may better fit the relatively small firm or short-term data.
- In the original PIN model, the point with 0 mostly be in the cluster of "no news". We think that 0 could also represent news in the market, but traders tend not to trend.

$$P\{X=x\}=\begin{cases}\phi+(1-\phi)e^{-\lambda}, & x=0,\\ (1-\phi)\frac{\lambda^x e^{-\lambda}}{x!}, & x\geq 1,\end{cases}$$

Stock NTE in 2008



(a) 3 Normal Copulas

(b) 1 Gumbel + 2 Normal Copulas

Figure: Poisson rate of the NTE stock in 2008 under different copula combinations. All lines connect 3 points of the fit in one day.

Stock NTE in 2008

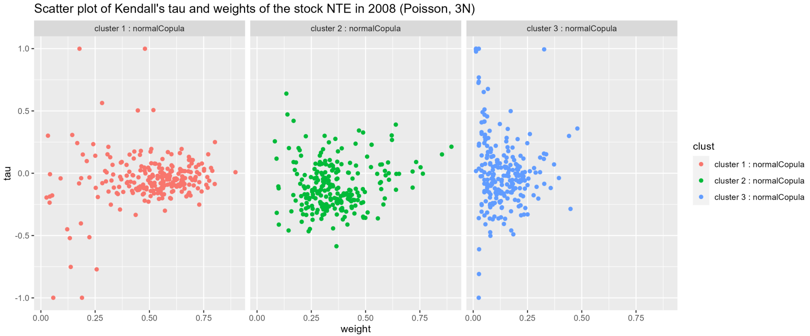


Figure: Scatter plot of Kendall's tau and the weights of the NTE stock in 2008 (Poisson, 3 normal).

Stock NTE in 2008

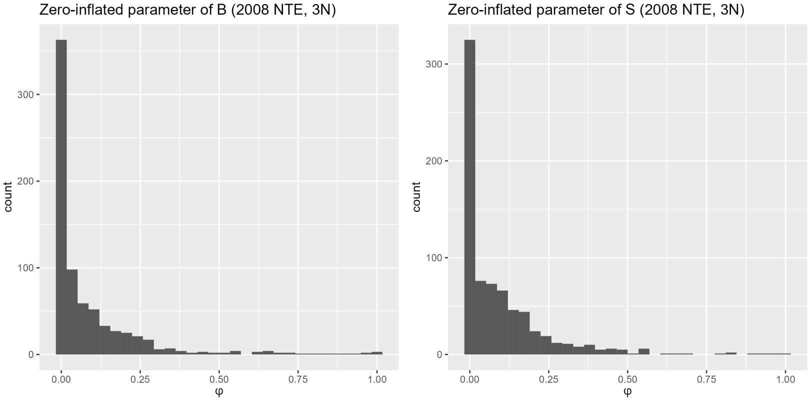


Figure: Histogram of zero-inflated parameters of the NTE stock in 2008 (3N)

Stock NTE in 2008/05/29

- We pick one day (2008/05/29) of the stock NTE and then check the cluster result and the marginal cdf.

Marginal	Copula	λ_B	λ_S	θ_j	τ_j	π_j	AIC
	normal	1.9244	1.6620	−0.1563	−0.0999	0.5250	
Poisson	normal	5.9193	12.3169	0.0196	0.0125	0.2777	925.6160
	normal	12.8452	4.2603	−0.0846	−0.0540	0.1973	
	normal	2.5152	3.4065	0.0623	0.0397	0.5315	
ZIP	normal	7.9531	13.7816	−0.2545	−0.1638	0.2346	883.4752
	normal	12.8518	3.7130	−0.1769	−0.1132	0.2339	
		ϕ_B	ϕ_S				
	normal	0.0607	0.3849				
ZIP	normal	0.0690	0.0010				
	normal	0.2777	0.0353				

Figure: The fitting result of the stock NTE (2008/05/29)

Stock NTE in 2008/05/29

(a) Poisson marginal (b) Zero-inflated Poisson marginal

Figure: Clustering result of the stock NTE in 2008/05/29 (3 normal)

Stock NTE in 2008/05/29

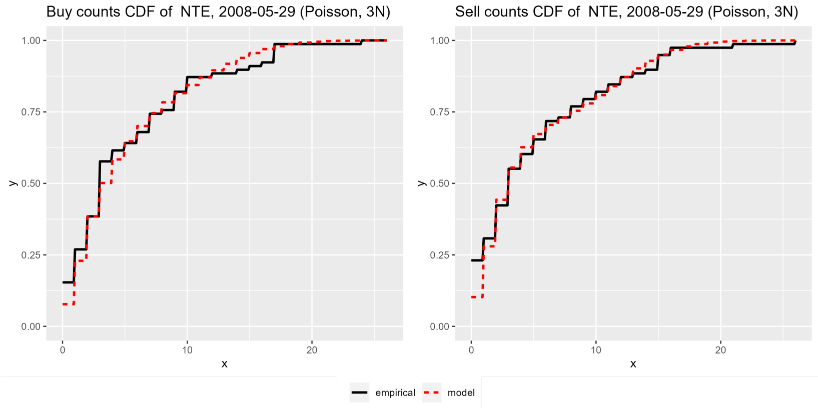


Figure: Marginal cdf of the NTE stock in 2008 (Poisson, 3 normal)

Stock NTE in 2008/05/29

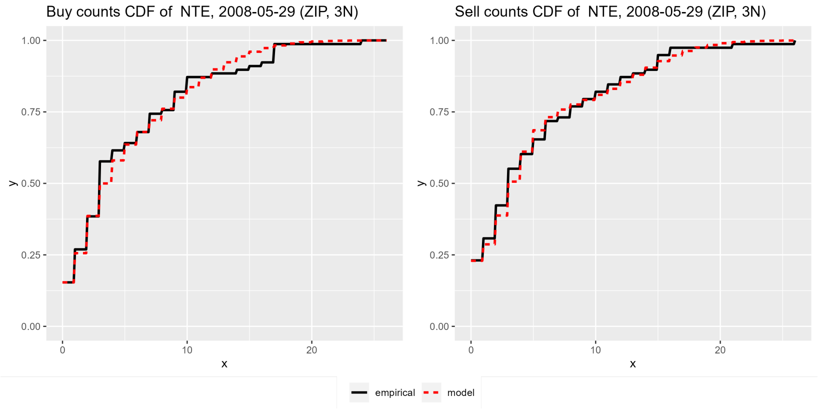


Figure: Marginal cdf of the NTE stock in 2008 (ZIP, 3 normal)

Stock NTE in 2008/05/29

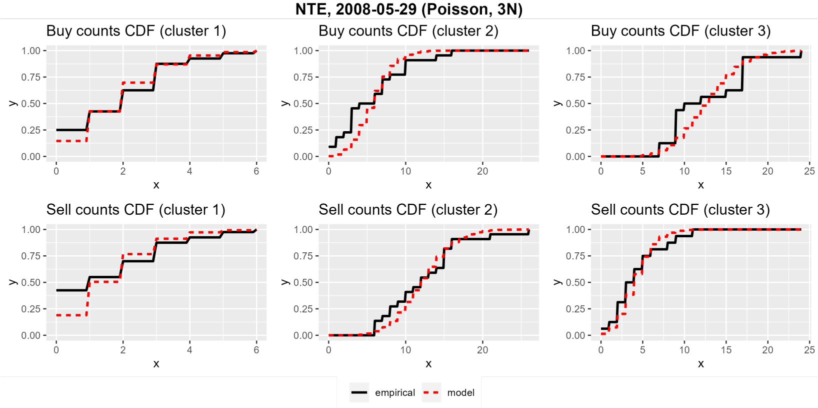


Figure: Marginal cdf of each cluster of the NTE stock in 2008 (Poisson, 3 normal)

Stock NTE in 2008/05/29

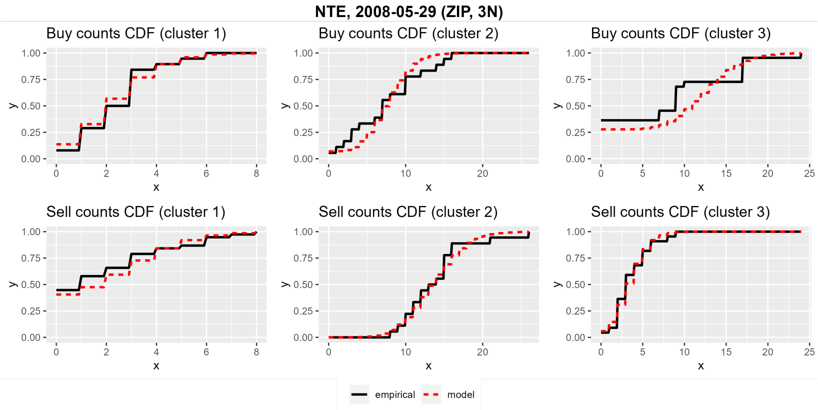


Figure: Marginal cdf of each cluster of the NTE stock in 2008 (ZIP, 3 normal)

Discussion and Future work

- Proposed a method (co-PIN) to evaluate the PIN model with the correlation structure by using the copula function and the ECM algorithm.
- We observed a correlation in the PIN model, proving that the independent assumption was too strict.
- In future work, we aim to modify the code to fit the rotate copula and find a suitable method to obtain initial parameters.
- From the empirical studies results, the issue of misspecification in the mixture copula model is also an interesting topic for further research.

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